Project Proposal

Dhaka Air Quality Prediction System

Web Platform with ML Prediction & Dataset Upload (working title)

Team members: _____

Course: CSE299

Date: _____

1. Introduction

Air pollution is a major environmental problem in Dhaka, but raw pollution measurements are difficult for many users to understand. This project proposes a web application that uses machine learning to analyze air-quality data, predict AQI or air-quality categories, compare model performance, and present results through a simple dashboard. The system will begin with a Mendeley air-quality dataset and will be designed so that new datasets can be uploaded and tested later.

2. Problem Statement

- Dhaka experiences frequent high levels of PM2.5 and PM10, but users often see only raw numbers without clear interpretation.
- Air-quality datasets can contain missing values, different formats, duplicates, and inconsistent columns, which makes analysis difficult.
- A model trained on one fixed dataset may become less useful when new data is available, so the system should support adding and testing new datasets.

3. Objectives

1. Build a web-based air-quality prediction and analytics system for Dhaka.
2. Use machine learning to predict AQI values or air-quality categories from pollution and environmental features.
3. Compare multiple ML models and identify the best-performing approach using clear evaluation metrics.
4. Allow users to upload a new CSV/Excel dataset, validate it, preprocess it, and use it for testing or retraining.
5. Present historical trends, predictions, model comparison, and what-if inputs through an easy-to-read dashboard.

4. Proposed Solution

The web application will load the initial Mendeley dataset, clean and preprocess the data, train multiple machine-learning models, and store evaluation results. A user can select historical data, request a prediction, compare models, or change input values such as PM2.5 and PM10 to observe how the predicted AQI changes. The system will also include a dataset-upload workflow that checks required columns, reports data-quality issues, and allows the new dataset to be analyzed without rebuilding the whole application.

5. Key Features

- **Air-quality dashboard:** PM2.5, PM10, AQI, trends, charts, and summary statistics.
- **ML prediction:** predict AQI or air-quality category using the selected model.
- **Model comparison:** compare multiple models using MAE/Accuracy and other suitable metrics.
- **Dataset upload:** add CSV/Excel files, validate columns, clean data, and run the pipeline on new data.
- **Historical data viewer:** filter and explore previous pollution measurements by date/time.
- **What-if testing:** change selected input values and observe how the model output changes.

6. System Architecture

The system will use one backend pipeline for data preparation, model training, prediction, and evaluation. The browser dashboard communicates with the backend through API routes. Uploaded datasets pass through validation and preprocessing before they are stored or sent to the ML pipeline. Model outputs and evaluation results are then returned to the dashboard for charts, comparisons, and predictions.

7. Technology Stack

Layer	Technology	Purpose
Language	Python	Data processing, machine learning, and backend logic
Frontend	HTML, CSS, JavaScript	Dashboard, filters, forms, and user interaction
Charts	Chart.js / Plotly	AQI trends, actual-vs-predicted charts, model comparison
Backend	Flask	API routes, file upload, prediction requests, model integration
Data processing	Pandas, NumPy	Cleaning, transformation, missing-value handling
ML	Scikit-learn	Training, testing, model comparison, evaluation metrics
Database / storage	SQLite / local structured storage	Dataset metadata, model results, upload history
Version control	Git + GitHub	Team collaboration, code history, issue tracking
Deployment	Render / Railway / similar	Hosting the final web application

8. Data & ML Plan

Initial dataset: CAMS-BD / Bangladesh air-quality data from Mendeley. The working features may include PM2.5, PM10, temperature, humidity, wind speed, date/time, and available pollutant variables. The exact target will be finalized after inspecting data quality: either numerical AQI prediction (regression) or air-quality category prediction (classification).

- **Preprocessing:** remove duplicates, handle missing values, standardize column names, encode time features, and split data into training and testing sets.
- **Model comparison:** start with a simple baseline and compare it with stronger ML models such as Random Forest and Gradient Boosting/XGBoost.
- **Evaluation:** use MAE/RMSE for numerical AQI prediction, or Accuracy/F1 Score for category prediction.
- **New-data support:** validate uploaded datasets against required fields and allow retraining or testing when the structure is compatible.

9. Development Plan

Phase	Work
Phase 1	Finalize requirements, inspect Mendeley dataset, define target and required columns
Phase 2	Clean data, exploratory analysis, feature preparation, and baseline model
Phase 3	Train and compare ML models; add evaluation and error analysis
Phase 4	Build dashboard, prediction form, charts, and historical data filters
Phase 5	Add dataset upload/validation, integrate backend, test full system, deploy demo

10. Risks and Mitigation

Risk	Mitigation
Missing or inconsistent dataset values	Use validation, cleaning rules, missing-value handling, and clear error messages.
Different uploaded dataset formats	Define required columns and map common alternate column names before processing.
Model overfitting or weak generalization	Use train/test split or cross-validation and compare multiple models.
Unbalanced air-quality categories	Use F1 score and class-aware evaluation, not accuracy alone.
New dataset performs differently	Report performance separately and retrain only when the uploaded data is compatible.
Time limit for a 3-member team	Keep live-data integration optional and prioritize the complete ML + dashboard pipeline first.

11. Evaluation Plan

The system will be evaluated at both model level and application level. For regression, lower MAE and RMSE indicate better AQI predictions. For classification, higher Accuracy and F1 Score indicate better category prediction. We will also compare models on the same test data, inspect common errors, and test whether the pipeline can accept a compatible new dataset without breaking the application.

12. Expected Outcome

A working web application that uses real air-quality data to train and compare machine-learning models, predicts AQI or air-quality categories, visualizes historical and predicted results, and supports adding new compatible datasets. The final submission will include the cleaned dataset, trained models, evaluation results, error analysis, dashboard, upload workflow, source code, and a live demo.

13. Week-by-Week Schedule

Week	Work
Week 3	Proposal presentation and final scope confirmation
Week 4	Dataset inspection, cleaning, and preprocessing
Week 5	Exploratory analysis and baseline model
Week 6	Train and compare ML models
Week 7	Mid-project report and evaluation review
Week 8	Backend API and prediction integration
Week 9	Dashboard, graphs, filters, and model comparison UI
Week 10	Dataset upload, validation, and new-data testing
Week 11	Full-system testing, error analysis, and deployment
Week 12	Final presentation, live demo, and final report